

WNC Salamanders - In-Class PDO Labs

Course: WEB-250

Format: Three ungraded, instructor-led labs

Target length: 15-20 minutes per lab, after the shared setup below

Project folder: `salamanders`

Database: `wnc_salamanders`

This is a companion project for **asgn07**, not a replacement for the WNC Bird Guide. All three labs build the same application in the same folder. Keep the work between class meetings; do not start a new project for each lab.

Work through these files in order	New focus	Result
<code>01_pdo_connection.md</code>	PDO connection inside the same <code>Database</code> class pattern used in <code>asgn07</code> .	A successful connection page.
<code>02_display_data.md</code>	<code>query()</code> , <code>fetchColumn()</code> , and <code>foreach</code> in an HTML table.	Database name and live record counts.
<code>03_error_handling.md</code>	<code>try</code> , <code>catch</code> , exceptions, and private error logging.	A useful response to connection or query failures.

These are practice activities: **no grade, rubric, upload, or formal submission**. Each ends with a short experiment and a verbal checkpoint. Keep the code ready for the next lab.

Scope

SQL is supplied because the new material is PHP's interaction with the database. We will not reteach table creation, joins, or relationship design. The practice schema has `salamanders`, `habitats`, and `salamander_habitat_links`. A fresh import contains **5 species records, 3 habitat records, and 9 links**.

The final browser page is a summary table, like `asgn07`, not an animal listing or gallery. No CSS, filters, callbacks, `array_filter()`, image processing, autoloading, or new species model classes are involved.

Credentials stay in a PHP configuration file excluded by `.gitignore`. **Using a `.env` file is reserved for a later assignment**. No environment-variable package is needed here.

Shared setup - complete before the first timed lab

Put the project in the web folder

Extract `salamanders_in_class_labs.zip`. Put the resulting `salamanders` folder alongside your other WEB-250 work. Typical locations are:

```
macOS / MAMP:  
/Applications/MAMP/htdocs/web250/salamanders
```

```
Windows / XAMPP:  
C:\xampp\htdocs\web250\salamanders
```

Use your established document root if it differs. Open **that folder** in VS Code. Do not accidentally nest it as **salamanders/salamanders**.

```
salamanders/  
|-- README-START-HERE.md  
|-- 01_pdo_connection.md  
|-- 02_display_data.md  
|-- 03_error_handling.md  
|-- DATA-NOTES.md  
|-- INSTRUCTOR-NOTES.md  
|-- .gitignore  
|-- private/  
|   |-- .htaccess  
|   |-- db_credentials.php.example  
|   |-- functions.php  
|   |-- initialize.php  
|   |-- classes/  
|       |-- Database.php  
|-- public/  
|   |-- index.php  
|-- sql/  
|   |-- .htaccess  
|   |-- wnc_salamanders.sql
```

The starter class is intentionally unfinished. The starter index page simply confirms that your browser is reaching this project. You will create **private/db_credentials.php** during Lab 01.

Start the established servers

Use **MAMP or XAMPP** to start both the web server and MySQL/MariaDB. Open the starter page through **http://localhost**, not by double-clicking the PHP file and not through VS Code Live Server.

Examples, using the common web ports:

```
MAMP with Apache on port 8888:  
http://localhost:8888/web250/salamanders/public/index.php
```

```
XAMPP with Apache on port 80:  
http://localhost/web250/salamanders/public/index.php
```

```
An existing class setup with Apache on port 55000:  
http://localhost:55000/web250/salamanders/public/index.php
```

Use **your actual Apache port** in the URL. MAMP's default web and MySQL ports are different; verify both in its settings. [MAMP ports](#) and [XAMPP for Windows](#) document the local-server setup.

Checkpoint: The browser shows [Salamanders starter page](#).

Import the separate practice database

Open phpMyAdmin or your established SQL tool. Import `sql/wnc_salamanders.sql`. The script creates and selects `wnc_salamanders`; it does not change `wnc_birds`.

For a new practice database, the final query should show:

table_name	row_count
salamanders	5
habitats	3
salamander_habitat_links	9

The script can be re-imported into the same compatible schema without duplicating its seeds. It preserves existing records and edits; it is not a reset script or a migration for incompatible tables. Stop on any import error rather than dropping an existing database or disabling foreign-key checks. Extra records will produce larger counts.

The links use a composite primary key and cascading updates/deletes on both foreign keys. As with `asgn07`, deleting a parent removes its links, not the other parent records. There is no need to practice deletion in these labs.

Keep credentials local

Keep the supplied `.gitignore`. Its `/private/db_credentials.php` rule excludes the local credentials file when it is untracked. An ignore rule does **not** remove a file already tracked by Git, protect a published website, or exclude a file from a ZIP. Do not share real credentials. [Git ignore reference](#).

The `.htaccess` files deny direct Apache access to `private/` and `sql/` where overrides are supported. PHP can still load the files by their filesystem paths. These are local class projects, not deployment instructions; the directory name `private` alone is not access control.

How the starter files help

`private/initialize.php` loads the credentials, `h()` helper, and `Database` class with explicit `require_once` calls. It deliberately does **not** connect. Lab 03 will put the connection call in the page's `try` block.

`private/functions.php` supplies the same small output-escaping helper used in `asgn07`. Students inspect it in Lab 02 rather than creating another helper from scratch.

Only `Database.php`, `index.php`, and the student's local credentials file need editing across the three labs. The Markdown files contain the finished code for each step; the ZIP starts at **the beginning of Lab 01**, not the end of Lab 03.

Before Lab 03

Have the instructor point out the local **PHP error log** so that finding the log does not consume the activity. On MAMP for macOS it is normally `/Applications/MAMP/logs/php_error.log`. XAMPP's location depends on the PHP configuration serving the browser. Lab 03 includes a local-only check for the effective setting. [MAMP log locations](#).

Instructor pacing note

The 15-20 minute targets assume the SQL import and starter-page check are already complete. Use the code blocks for guided typing or copying, then spend the remaining minutes on the experiment and explanation. There is intentionally no separate SQL-design exercise or graded reflection.

`asgn07` itself has not been rewritten by this package. These labs introduce the PHP techniques students will use there.